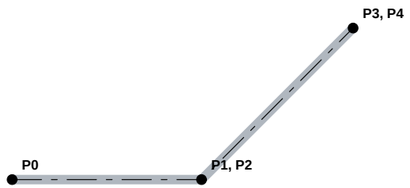
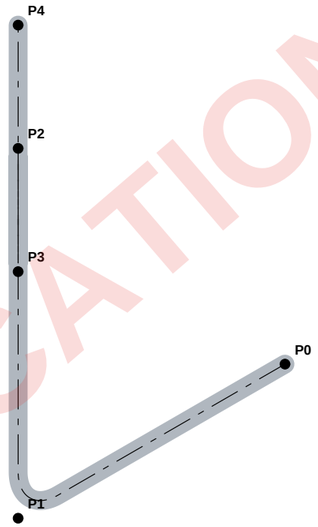
NOT FOR FABRICATION:

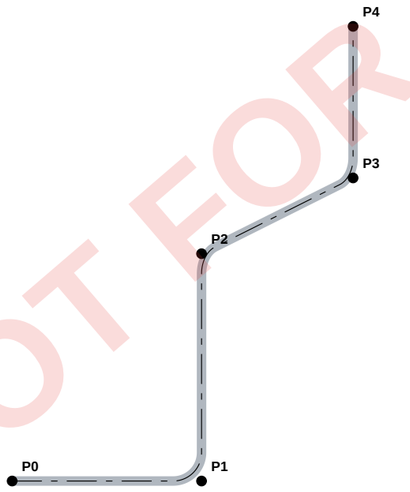
- 8 unverified or missing data value(s): Material specification, Minimum CLR, Minimum straight between bends, End 1 allowance ...



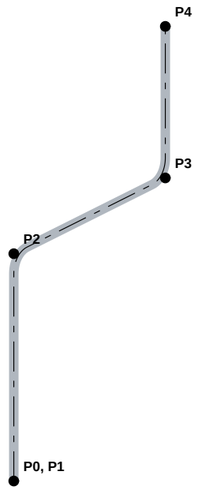
TOP VIEW
SCALE 1:4



ISOMETRIC (REFERENCE)
NOT TO SCALE



FRONT VIEW
SCALE 1:4



RIGHT SIDE VIEW
SCALE 1:4

BEND TABLE (INCHES, DEGREES). MARKS: DEVELOPED DISTANCE FROM P0.

PT	X	Y	Z	CLR	BEND °	ROT °	STRAIGHT TO NEXT	BEND MARKS
P0	0.0000	0.0000	0.0000				4.2500	
P1	5.0000	0.0000	0.0000	0.7500	90.00	90.00	4.7197	4.2500 - 5.4281
P2	5.0000	0.0000	6.0000	0.7500	70.53	-135.00	4.9393	10.1478 - 11.0710
P3	9.0000	4.0000	8.0000	0.7500	70.53	180.00	3.4697	16.0103 - 16.9335
P4	9.0000	4.0000	12.0000					

REVISIONS

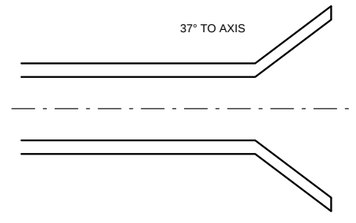
REV	DESCRIPTION	DATE	BY

NOTES:

1. MATERIAL: 304 CRES, SPEC NOT ENTERED.
2. TUBE: OD 0.2500 X WALL 0.0350 INCHES (-4).
3. DEVELOPED LENGTH 20.4032; CUT LENGTH 20.4032 - INCOMPLETE, END ALLOWANCE NOT ENTERED.
4. END 1: 37° SINGLE FLARE; END 2: 37° SINGLE FLARE. END POINTS SIT AT THE FLARE: NOT ENTERED.
5. STANDARDS AND DATA USED: SAE AS33611, REV ? - UNVERIFIED/NOT ENTERED; SHOP DATA, REV ? - UNVERIFIED/NOT ENTERED; SAE AS4330, REV ? - UNVERIFIED/NOT ENTERED; SAE AS4330 DATA FILE, TABLE NOTE - UNVERIFIED/NOT ENTERED.
6. Z UP. P0 = END 1. ROTATION + = CW LOOKING ALONG THE TUBE TOWARD END 2. FEED = STRAIGHT BETWEEN TANGENT POINTS. CLR = CENTERLINE BEND RADIUS. B1 FROM REFERENCE PLANE (+Z PROJECTED).
7. GENERAL TOLERANCES: NOT SPECIFIED.
8. CLEANLINESS: NOT SPECIFIED.
9. PROOF TEST: NOT SPECIFIED.
10. APPROVED DEVIATIONS: NONE.
11. DATA SET 0.1.0-placeholder, SHA-256 fbc06821c09ef860.
12. ALL DIMENSIONS PER BEND TABLE. EVERY VALUE IS LISTED WITH ITS FORMULA AND INPUTS IN THE CALCULATION RECORD (SHEET 2 ON).
13. DESIGN AID ONLY. ALL GEOMETRY MUST BE VERIFIED AND APPROVED PER YOUR ORGANIZATION'S ENGINEERING AND QUALITY PROCESS BEFORE FABRICATION.

DETAIL END 1: 37° FLARE

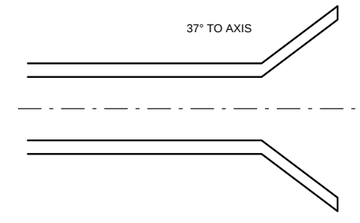
PER SAE AS4330 REV ?, TABLE ? - UNVERIFIED
FLARE DIA NOT ENTERED - NOT ENTERED
FLARE ANGLE NOT ENTERED ± NOT ENTERED
NUT AND SLEEVE: REFERENCE ONLY (SAE J514)



SCHEMATIC HALF-SECTION - NOT TO SCALE

DETAIL END 2: 37° FLARE

PER SAE AS4330 REV ?, TABLE ? - UNVERIFIED
FLARE DIA NOT ENTERED - NOT ENTERED
FLARE ANGLE NOT ENTERED ± NOT ENTERED
NUT AND SLEEVE: REFERENCE ONLY (SAE J514)



SCHEMATIC HALF-SECTION - NOT TO SCALE

TITLE
3D COMPOUND BEND TO PRESSURE PANEL

PART NUMBER	REV	SIZE
SAMPLE-3	**-**	**B**

DRAWN	CHECKED	DATE
		2026-10-18

SCALE	UNITS	SHEET
1:4	**INCHES**	**1 OF 3**

ENGINEERING APPROVAL	QUALITY APPROVAL

NOT FOR FABRICATION

CALCULATION RECORD

KEY	DESCRIPTION	FORMULA / STATUS	VALUE, INPUTS, SOURCE
resolved:od	Tube OD	INPUT [DERIVED]	0.25 in source: (no source), rev ?, basis: Dash-number definition: OD = dash / 16 in [tube_sizes row 0: dash=-4]
resolved:material_spec	Material specification	INPUT [MISSING]	— source: (no source), rev ? (UNVERIFIED – REPLACE FROM STANDARD; value not entered) [materials row 0: material_id=cres304]
resolved:wall	Wall	INPUT [USER INPUT]	0.035 in source: user input: wall entered in project
resolved:min_clr	Minimum CLR	INPUT [MISSING]	— source: SAE AS33611, rev ? (UNVERIFIED – REPLACE FROM STANDARD; value not entered) [as33611_bend_radius row 0: dash=-4, material_id=cres304, wall=any]
resolved:min_straight_between_bends	Minimum straight between bends	INPUT [MISSING]	— source: Shop data, rev ? (UNVERIFIED – REPLACE FROM SHOP DATA; value not entered) [shop_limits row 0: dash=-4, material_id=any]
resolved:end1.allowance	End 1 allowance	INPUT [MISSING]	— source: SAE AS4330, rev ? (UNVERIFIED – REPLACE FROM STANDARD; value not entered) [as4330_flare row 0: dash=-4, material_id=any, wall=any]
resolved:end1.min_straight	End 1 minimum straight	INPUT [MISSING]	— source: Shop data, rev ? (UNVERIFIED – REPLACE FROM SHOP DATA; value not entered) [shop_limits row 0: dash=-4, material_id=any]
resolved:end2.allowance	End 2 allowance	INPUT [MISSING]	— source: SAE AS4330, rev ? (UNVERIFIED – REPLACE FROM STANDARD; value not entered) [as4330_flare row 0: dash=-4, material_id=any, wall=any]
resolved:end2.min_straight	End 2 minimum straight	INPUT [MISSING]	— source: Shop data, rev ? (UNVERIFIED – REPLACE FROM SHOP DATA; value not entered) [shop_limits row 0: dash=-4, material_id=any]
resolved:end_point_definition	End point definition (on flare)	INPUT [MISSING]	— source: SAE AS4330 data file, table note [as4330_flare meta]
resolved:arc_length_factor	Arc length factor	INPUT [DEFAULT]	1.0 source: Shop data, rev ?, basis: No correction: pure centerline development and no springback. Replace with test-bend results. [shop_bending row 0: material_id=cres304, dash=any]
resolved:per_bend_adjust	Per-bend length adjust	INPUT [DEFAULT]	0.0 in source: Shop data, rev ?, basis: No correction: pure centerline development and no springback. Replace with test-bend results. [shop_bending row 0: material_id=cres304, dash=any]
input:P0	Point P0	INPUT [USER INPUT]	0.0, 0.0, 0.0 in
input:P1	Point P1	INPUT [USER INPUT]	5.0, 0.0, 0.0 in
input:P2	Point P2	INPUT [USER INPUT]	5.0, 0.0, 6.0 in
input:P3	Point P3	INPUT [USER INPUT]	9.0, 4.0, 8.0 in
input:P4	Point P4	INPUT [USER INPUT]	9.0, 4.0, 12.0 in
input:clr1	Bend 1 CLR	INPUT [USER INPUT]	0.75 in
input:clr2	Bend 2 CLR	INPUT [USER INPUT]	0.75 in
input:clr3	Bend 3 CLR	INPUT [USER INPUT]	0.75 in
seg0.leg	Leg 0 intersection-to-intersection length	L = |P[j+1] - P[j]|	= 5.0 in with P[j]=(0, 0, 0), P[j+1]=(5, 0, 0)
seg1.leg	Leg 1 intersection-to-intersection length	L = |P[j+1] - P[j]|	= 6.0 in with P[j]=(5, 0, 0), P[j+1]=(5, 0, 6)
seg2.leg	Leg 2 intersection-to-intersection length	L = |P[j+1] - P[j]|	= 6.0 in with P[j]=(5, 0, 6), P[j+1]=(9, 4, 8)
seg3.leg	Leg 3 intersection-to-intersection length	L = |P[j+1] - P[j]|	= 4.0 in with P[j]=(9, 4, 8), P[j+1]=(9, 4, 12)
bend1.angle	Bend 1 angle	theta = atan2(|u x v|, u . v)	= 1.5707963267948966 rad with u=(1, 0, 0), v=(0, 0, 1)
bend1.angle_deg	Bend 1 angle in degrees	theta_deg = theta * 180 / pi	= 90.0 deg with theta=1.5707963267948966
bend2.angle	Bend 2 angle	theta = atan2(|u x v|, u . v)	= 1.2309594173407747 rad with u=(0, 0, 1), v=(0.666667, 0.666667, 0.333333)
bend2.angle_deg	Bend 2 angle in degrees	theta_deg = theta * 180 / pi	= 70.52877936550931 deg with theta=1.2309594173407747
bend3.angle	Bend 3 angle	theta = atan2(|u x v|, u . v)	= 1.2309594173407747 rad with u=(0.666667, 0.666667, 0.333333), v=(0, 0, 1)
bend3.angle_deg	Bend 3 angle in degrees	theta_deg = theta * 180 / pi	= 70.52877936550931 deg with theta=1.2309594173407747
bend1.tangent	Bend 1 tangent length (vertex to tangent point)	T = R * tan(theta / 2)	= 0.7499999999999999 in with R=0.75, theta=1.5707963267948966
bend2.tangent	Bend 2 tangent length (vertex to tangent point)	T = R * tan(theta / 2)	= 0.5303300858899107 in with R=0.75, theta=1.2309594173407747
bend3.tangent	Bend 3 tangent length (vertex to tangent point)	T = R * tan(theta / 2)	= 0.5303300858899107 in with R=0.75, theta=1.2309594173407747
seg0.straight	Leg 0 straight between tangent points	S = L - T_start - T_end	= 4.25 in with L=5.0, T_start=0.0, T_end=0.7499999999999999
seg1.straight	Leg 1 straight between tangent points	S = L - T_start - T_end	= 4.7196699141100895 in with L=6.0, T_start=0.7499999999999999, T_end=0.5303300858899107
seg2.straight	Leg 2 straight between tangent points	S = L - T_start - T_end	= 4.939339828220179 in with L=6.0, T_start=0.5303300858899107, T_end=0.5303300858899107
seg3.straight	Leg 3 straight between tangent points	S = L - T_start - T_end	= 3.4696699141100895 in with L=4.0, T_start=0.5303300858899107, T_end=0.0
bend1.arc	Bend 1 nominal centerline arc	A = R * theta	= 1.1780972450961724 in with R=0.75, theta=1.5707963267948966
bend1.arc_dev	Bend 1 developed arc length	A_dev = A * f + adj	= 1.1780972450961724 in with A=1.1780972450961724, f=1.0, adj=0.0
bend1.rotation_from_ref_deg	Bend 1 plane rotation from the reference plane (+Z projected normal to the first leg; +Y if that leg is vertical)	B1 = atan2((n_ref x n) . d0, n_ref . n) * 180 / pi	= 90.0 deg with n_ref=(0, 0, 1), n=(0, -1, 0), d0=(1, 0, 0)
bend1.mark_start	Bend 1 start mark from end 1 tube end	M_start = sum(S before) + sum(A_dev before)	= 4.25 in with value=4.25
bend1.mark_end	Bend 1 end mark from end 1 tube end	M_end = M_start + A_dev	= 5.428097245096172 in with M_start=4.25, A_dev=1.1780972450961724
bend2.arc	Bend 2 nominal centerline arc	A = R * theta	= 0.9232195630055811 in with R=0.75, theta=1.2309594173407747
bend2.arc_dev	Bend 2 developed arc length	A_dev = A * f + adj	= 0.9232195630055811 in with A=0.9232195630055811, f=1.0, adj=0.0
bend2.rotation	Bend 2 plane rotation from previous bend (+ = clockwise looking toward end 2)	B = atan2((n_prev x n) . d, n_prev . n)	= -2.356194490192345 rad with n_prev=(0, -1, 0), n=(-0.707107, 0.707107, 0), d=(0, 0, 1)
bend2.rotation_deg	Bend 2 plane rotation in degrees	B_deg = B * 180 / pi	= -135.0 deg with B=-2.356194490192345
bend2.mark_start	Bend 2 start mark from end 1 tube end	M_start = sum(S before) + sum(A_dev before)	= 10.147767159206262 in with value=10.147767159206262
bend2.mark_end	Bend 2 end mark from end 1 tube end	M_end = M_start + A_dev	= 11.070986722211844 in with M_start=10.147767159206262, A_dev=0.9232195630055811
bend3.arc	Bend 3 nominal centerline arc	A = R * theta	= 0.9232195630055811 in with R=0.75, theta=1.2309594173407747
bend3.arc_dev	Bend 3 developed arc length	A_dev = A * f + adj	= 0.9232195630055811 in with A=0.9232195630055811, f=1.0, adj=0.0
bend3.rotation	Bend 3 plane rotation from previous bend (+ = clockwise looking toward end 2)	B = atan2((n_prev x n) . d, n_prev . n)	= 3.141592653589793 rad with n_prev=(-0.707107, 0.707107, 0), n=(0.707107, -0.707107, 0), d=(0.666667, 0.666667, 0.333333)

CALCULATION RECORD

KEY	DESCRIPTION	FORMULA / STATUS	VALUE, INPUTS, SOURCE
bend3.rotation_deg	Bend 3 plane rotation in degrees	B_deg = B * 180 / pi	= 180.0 deg with B=3.141592653589793
bend3.mark_start	Bend 3 start mark from end 1 tube end	M_start = sum(S before) + sum(A_dev before)	= 16.010326550432023 in with value=16.010326550432023
bend3.mark_end	Bend 3 end mark from end 1 tube end	M_end = M_start + A_dev	= 16.933546113437604 in with M_start=16.010326550432023, A_dev=0.9232195630055811
total.centerline	Centerline length (tangent straights + nominal arcs)	C = sum(S) + sum(A)	= 20.403216027547693 in with sum(S)=17.378679656440358, sum(A)=3.0245363711073345
total.developed	Developed length (with shop arc factor / per-bend adjust)	D = sum(S) + sum(A_dev)	= 20.403216027547693 in with sum(S)=17.378679656440358, sum(A_dev)=3.0245363711073345
end1.allowance	End 1 end-prep / flare allowance	input	= 0.0 in
end2.allowance	End 2 end-prep / flare allowance	input	= 0.0 in
total.cut	Cut (stock) length	Cut = D + FA1 + FA2	= 20.403216027547693 in with D=20.403216027547693, FA1=0.0, FA2=0.0
bend1.wall_extrados_est	Estimated extrados wall (simple strain model, screening only)	t_out ~= t * 2R / (2R + OD)	= 0.030000000000000002 in with t=0.035, R=0.75, OD=0.25
bend2.wall_extrados_est	Estimated extrados wall (simple strain model, screening only)	t_out ~= t * 2R / (2R + OD)	= 0.030000000000000002 in with t=0.035, R=0.75, OD=0.25
bend3.wall_extrados_est	Estimated extrados wall (simple strain model, screening only)	t_out ~= t * 2R / (2R + OD)	= 0.030000000000000002 in with t=0.035, R=0.75, OD=0.25

Values at full double precision as stored; drawing sheet values are these rounded for display (4 dp lengths, 2 dp angles).